Used Car Price Prediction

Machine Learning Project Proposal

1. Problem Statement

Estimating the selling price of a used car is challenging because prices depend on several factors, such as the car's age, present price, kilometers driven, fuel type, and transmission. This project uses Machine Learning regression models to predict used car selling prices based on these features, helping users obtain a price estimate and make more informed decisions.

2. Target Users and Importance

The project is designed for used car buyers and sellers who want a quick estimate of a car's selling price based on its features. It can help buyers compare prices, support sellers in setting reasonable asking prices, and make the pricing process easier. Predictions are based on an Indian used-car dataset, so they may not reflect prices in other markets.

3. Dataset

Dataset: CarDekho Used Car Dataset
Source: Kaggle — vehicle-dataset-from-cardekho
Dataset link: https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho
Original size: 301 rows and 9 columns
After duplicate removal: 299 rows and 9 columns
Target variable: Selling_Price (in lakh Indian rupees)

4. Project Objectives

• Inspect and clean the dataset, including checking missing values, duplicates, data types, and suspicious values.

• Use Exploratory Data Analysis (EDA) to understand how car features relate to selling price.

• Train and compare regression algorithms, including Linear Regression, Decision Tree, and Random Forest.

• Evaluate models using MAE, RMSE, and R^2, and tune model hyperparameters where appropriate.

• Save the selected model and create a Streamlit web application for interactive price estimates.

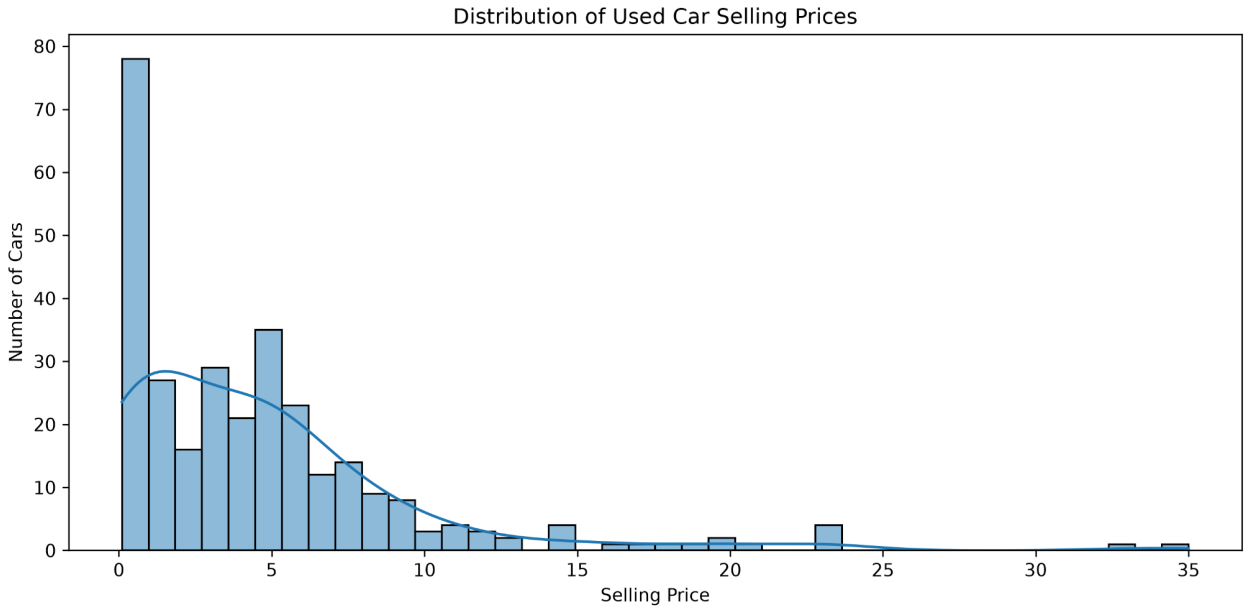
5. Methodology

Step 1 — Data Collection: Load the CarDekho dataset.
Step 2 — Preprocessing: Inspect the data, remove duplicate records, check missing values, and prepare features.
Step 3 — EDA: Explore the target distribution, compare selling price with selected features, and inspect numeric correlations.
Step 4 — Model Training: Split the data into training and testing sets and train several regression models.
Step 5 — Evaluation and Tuning: Compare test-set metrics and use cross-validation and hyperparameter search where useful.
Step 6 — Deployment: Save the chosen model and connect it to a Streamlit interface.

6. Initial Exploratory Data Analysis

The following plots are the initial visual analysis for the project. They help identify patterns and guide feature selection and model evaluation. Correlation describes linear association and should not be interpreted as proof of causation.

Figure 1. Distribution of Used Car Selling Prices



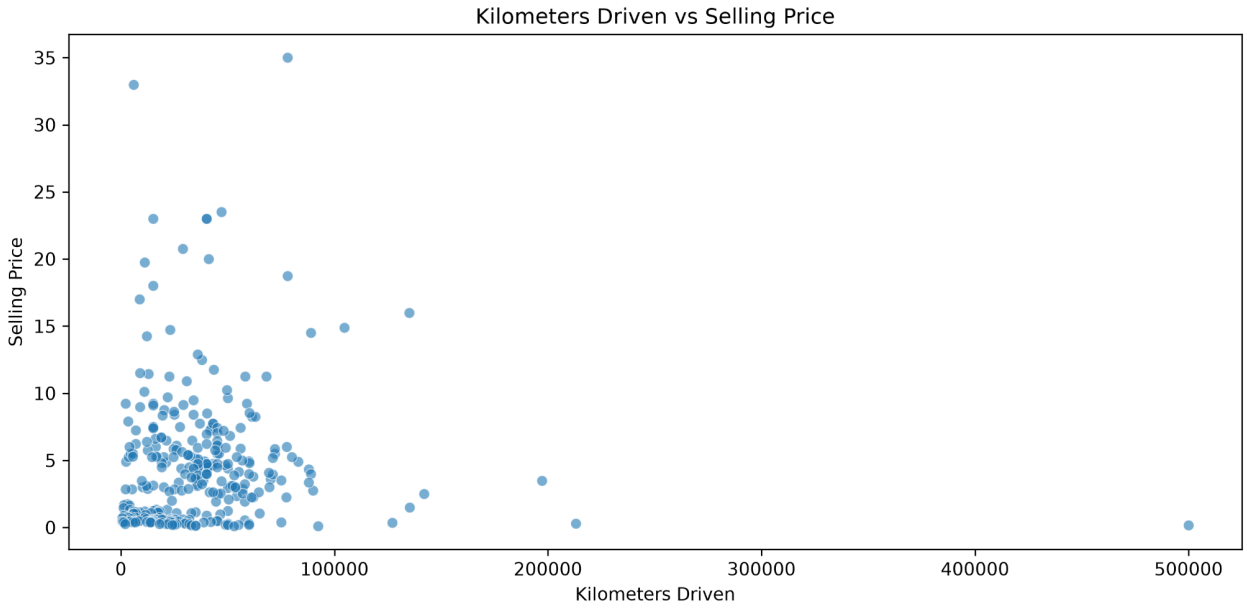
Most cars are concentrated at lower selling prices, with a right-skewed distribution and a small number of expensive cars.

Figure 2. Manufacturing Year vs Selling Price



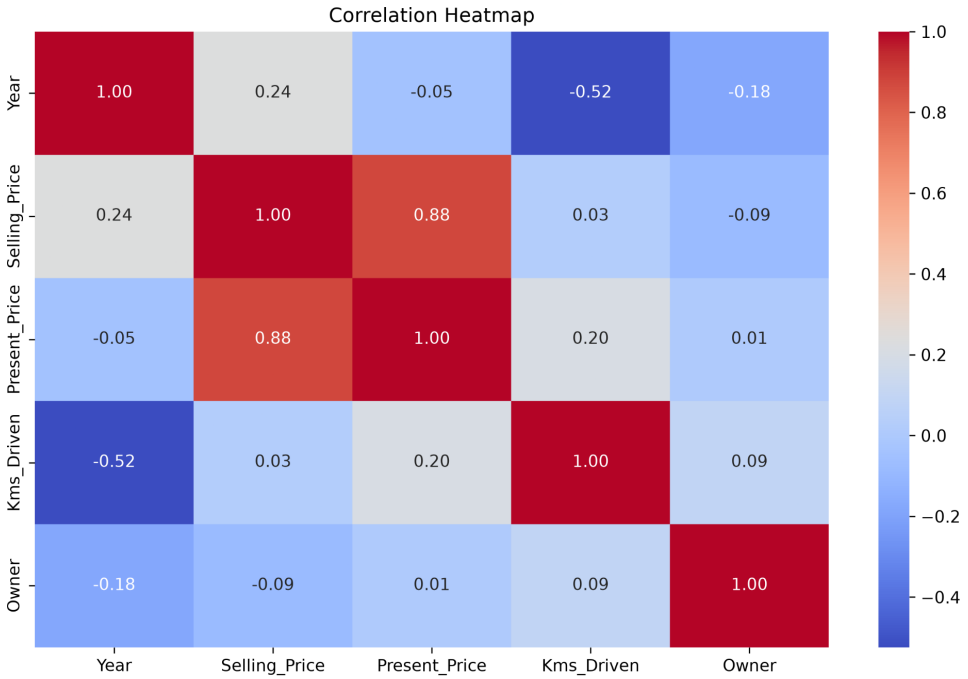
Newer cars generally reach higher selling prices, although prices vary considerably between individual cars.

Figure 3. Kilometers Driven vs Selling Price



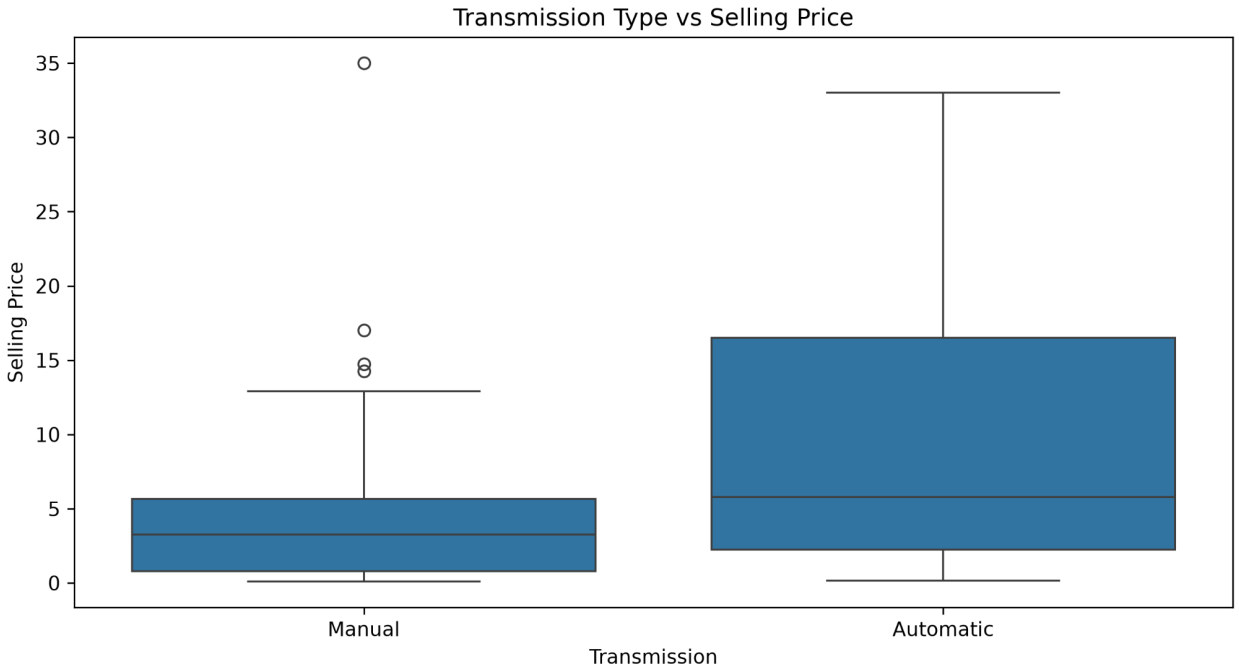
The relationship between kilometers driven and selling price is not strictly linear; the dataset also contains a few high-mileage observations.

Figure 4. Correlation Heatmap



Present price has a strong positive correlation with selling price (about 0.88). Year has a weaker positive correlation, while kilometers driven has little linear correlation in this dataset.

Figure 5. Transmission Type vs Selling Price



Automatic cars show a higher and wider selling-price distribution in this dataset. This is an observed association and does not by itself prove that transmission causes higher prices.

7. Expected Deliverables

The final deliverables include the cleaned dataset, preprocessing and EDA notebooks, trained regression models, model-comparison and evaluation results, a saved model file, project documentation, and a deployed Streamlit application.

8. Limitations

The dataset is relatively small and represents the Indian used-car market. Results may not generalize to other countries, market conditions, or car categories. The app's estimates are educational predictions rather than professional valuations or guaranteed market prices.

9. Project Links

GitHub repository: https://github.com/AbdlerahmanGihad/Used-Car-Price-Prediction
Live Streamlit application: https://used-car-price-prediction-a5hgsy3jmwmyihcwjqy4mt.streamlit.app/

Individual project — data collection, preprocessing, analysis, modeling, evaluation, deployment, and documentation are handled by the project author.